DPP DEPARTMENTAL POLICY AND PROCEDURE

TITLE:	RECOGNIZING CRITICAL ECG CHANGES
DEPARTMENT:	INTENSIVE CARE UNIT
APPLIES TO:	ALL MEDICAL AND NURSING STAFF

CODE AND VERSION NUMBER	
DPP-ICU-0007(v5)	
LAST VERSION APPROVAL DATE:	18 / 03 / 2022
NUMBER OF PAGES:	5

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APPROVED DATE	EFFECTIVE DATE	TYPE OF REVISION	REVIEW DATE
09 / 02 / 2025	23 / 02 / 2025	TRIENNIAL	08 / 02 / 2028

CONTENTS

1. PURPOSE	4. POLICY	7.REFERENCE/CROSS-REFERENCES
2. DEFINITION & ABBREVIATION	5. PROCEDURE	8. ATTACHMENT
3. RESPONSIBILITY	6. MEASUREMENT	9. REVISION CLAUSE

1. PURPOSE

1.1. To identify common abnormal cardiac rhythm.

1.2. To monitor the severity and progression of the arrhythmias noted.

1.3. To set guidelines on the process of identification, monitoring and reporting of critical ECG changes.

2. DEFINITION & ABBREVIATION

2.1. **Arrhythmias** - are variations in the normal pattern of electrical stimulation of the heart.

2.2. **ICU** - Intensive Care Unit

2.3. **Carotid Sinus Massage** - involves rubbing the large part of the arterial wall at the point where the common carotid artery, located in the neck, divides into its two main branches.

2.4. **Sinus tachycardia** - a sinus rhythm with an elevated rate of impulses greater than 100 beats per minute and usually up to 160 beats per minute.

2.4.1. Rate: greater than 100 per minute (usually up to 160)

2.4.2. Rhythm: regular

2.4.3. P waves = normal and upright; one P wave before each QRS and maybe buried in the T wave

2.4.4. P-R interval: normal (0.12 to 0.20 second) and constant

2.4.5. QRS Complex: normal (less than 0.12 second)

2.5. **Sinus Bradycardia** - a sinus rhythm with a rate that is lower than normal rate under 60 beats per minute.

2.5.1. Rate: less than 60 per minute

2.5.2. Rhythm: regular

2.5.3. P waves: normal and upright; one P wave before each QRS

2.5.4. P-R interval = normal (0.12 to 0.20 second) and constant

2.5.5. QRS Complex: normal (less than 0.12 second)

2.6. **Atrial Flutter** is a common abnormal heart rhythm that starts in the atrial chambers of the heart. It is usually associated with fast heart rate, more than 100 beats per minute.

2.6.1. Rate: atrial; 250 to 350 per minute, ventricular; 140 to 160 if untreated and 100 in treated patients

2.6.2. Rhythm: atrial; regular, ventricular; usually regular but maybe irregular

2.6.3. P waves: no true P waves, flutter waves resemble 'saw-tooth" pattern

2.6.4. P-R interval = usually constant but may vary

2.6.5. QRS Complex: usually normal

2.7. **Atrial Fibrillation** is a quivering or irregular heartbeat that can lead to blood clots, stroke, heart failure and other heart-related complications.

2.7.1. Rate: atrial; 400 to 700 per minute, ventricular; 160 to 180 if untreated and 100 in digitalized patients

2.7.2. Rhythm = irregularly irregular

2.7.3. P waves: no P waves, fibrillatory waves instead

2.7.4. P-R interval: none

2.7.5. QRS Complex: normal unless aberrant conduction occurs

2.8. **Ventricular Tachycardia (V-Tach or VT)** is a type of regular and fast heart rate that arises from improper electrical activity in the ventricles of the heart.

2.8.1. Rate = 100 to 220 per minute

2.8.2. Rhythm: usually regular

2.8.3. P waves: none associated with QRS complex

2.8.4. P-R interval: none

2.8.5. QRS Complex: width is 0.12 second or greater, bizarre in shape, ST segment and T wave are commonly opposite to the QRS.

2.9. **Ventricular Fibrillation (VF or VFib)** is a heart rhythm problem that occurs when the heart beats with rapid, erratic electrical impulses. This causes the pumping chambers; the ventricles, to quiver uselessly, instead of pumping blood.

2.9.1. Rate: 150 to 300 uncoordinated waves per minute

2.9.2. Rhythm: totally irregular

2.9.3. P waves: absent

2.9.4. P-R interval: none

2.9.5. QRS Complex: QRS complexes are absent, instead fibrillatory waves.

3. RESPONSIBILITY

3.1. **ICU Nurse** - notify physician once observed and document the arrhythmia and the treatment administered. identify cardiac

3.2. **ICU Head Nurse -** ensures that all staff is aware of this policy and supervises its implementation

3.3. **ICU Doctor** - verifies reported arrhythmia and manages accordingly.

4. POLICY

4.1. Patients admitted in ICU must be on continuous cardiac monitoring.

4.2. ICU nurse must be competent in identifying abnormal rhythms and reporting it promptly.

4.3. Nurse must be knowledgeable concerning the proper placement of the monitor leads including that of the defibrillator

4.4. ECG should be taken when doubtful cardiac rhythm noted in the cardiac monitor and if reverted to

sinus rhythm

4.5. Proper Maintenance of the machine must be employed to avoid malfunction.

4.6. In case of malfunction, the Biomedical Engineer must be called immediately for repair.

5. PROCEDURE

5.1. Ensure that patient is properly assessed and identified correctly. (Refer to APP-QPS-0016 Patient Identification).

5.2. Obtain patient past medical history for any arrhythmias recorded.

5.3. Place electrodes on the chest to monitor patients Electrocardiogram continuously.

5.3.1. **Cardiac Monitor**

5.3.1.1. Red on the right upper chest wall just below the clavicle

5.3.1.2. Yellow on the left upper chest wall just below the clavicle

5.3.1.3. White on the fourth intercostal space immediately right of sternum

5.3.1.4. Black on the right lower edge of the rib cage

5.3.1.5. Green on the left lower edge of the rib cage

5.3.2. **Defibrillator (LifePak)**

5.3.2.1. White on the right upper chest wall just below the clavicle

5.3.2.2. Black on the left upper chest wall just below the clavicle

5.3.2.3. Red on the left lower edge of the rib cage

5.3.3. **Defibrillator (Zoll)**

5.3.3.1. Red on the right upper chest wall just below the clavicle

5.3.3.2. Yellow on the left upper chest wall just below the clavicle

5.3.3.3. Green on the left lower edge of the rib cage

5.4. If any abnormal rhythm is observed;

5.4.1. Re-check placement of the electrodes.

5.4.2. Do ECG for proper identification of arrhythmia.

5.4.3. When certain about the arrhythmia notify the physician immediately.

5.4.4. Manage pharmacologically according to abnormal rhythm detected. Please refer to policy on Treatment of Arrhythmias. (Refer to DPP-ICU-0021 Management of Arrhythmias).

5.4.5. Document findings, the treatment given and patient response in the progress notes in the MCC.

6. MEASUREMENT

6.1. 100% of the medical and nursing staff are well oriented and adheres to this policy.

7. REFERENCES /CROSS- REFERENCES

7.1. Advance Cardiac Life Support, Saudi Heart Association Book 2012

7.2. DPP-NR-NE-ICU-010.2 (v1): Recognizing Arrhythmias

7.3. APP-QPS-0016 Patient Identification

8. ATTACHMENT

8.1. Recognizing Critical ECG Changes Competency Check Off

9. REVISION CLAUSE

9.1. Changes to the hospital policy and procedure template based from the Al-Baha Health Cluster logo, font face and design